

Accessible Health Screening: A Portable System for Early Detection of Toxin Elevation Linked to Kidney Disorders

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ABSTRACT

The objective of this work is to serve as an initial screening for potential kidney diseases. It serves as an early indicator of increased toxin levels in the human body. A large portion of the population still lacks immediate and urgent access to diagnostic services. It often overlooks the timely care required to prevent such health risks. To address these challenges, a cost-effective, user-friendly, and accessible prototype for detecting toxins in urine samples has been developed, providing results faster than conventional diagnostic techniques. The system offers an essential preliminary assessment to help determine whether an individual should seek further medical evaluation. Its performance is comparable to standard clinical procedures, and the results have been validated by CDAC, Kolkata, in accordance with their established protocols. This initiative aims to strengthen early detection efforts and improve health outcomes,



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particularly for communities with limited access to healthcare resources.

1 Introduction

Toxins can have a huge impact on kidney function, which might result in acute kidney injury (AKI) or chronic kidney disease (CKD). The kidneys filter blood and excrete waste products or byproducts through urine. When toxins such as heavy metals (e.g., lead, cadmium), and uremic toxins, i.e., Urea, Creatinine & Albumin toxins, accumulate in the bloodstream, they become the most prominent reason for impairing nephron function -the basic filtering cells of the kidney. According to the World Health Organization (WHO), the presence of high levels of nephrotoxic substances is the culprit for approximately 20% of AKI cases globally [1]. Cadmium, for example, has been observed to reduce glomerular filtration rate (GFR), causing irreversible kidney damage by sustaining such conditions [2]. In addition, toxins such as aristolochic acid are linked to Balkan Endothelial Nephropathy, a form of CKD with a strong carcinogenic profile [3]. The concentration of protein-bound uremic toxins such as indoxyl sulphate and p-cresyl sulphate in CKD is more responsible for the progression of renal fibrosis and cardiovascular complications [4]. Hence, early detection of high toxin concentration is critical to prevent irreversible renal disorders.

Standard methods for finding out the toxin concentration in biological fluids like blood and urine involve a series of tests and processes. High-performance liquid chromatography (HPLC) and gas chromatography-mass spectrometry (GC-MS) are currently proven to be the most accurate due to their high sensitivity in detecting the presence of toxins. Enzyme-linked immunosorbent assay (ELISA) kits are also used widely for detecting specific protein biomarkers such as NGAL and Albumin, both of which are considered to provide early signs of kidney damage. For example, ELISA kits for human Albumin (ab108788) and NGAL (ab119600) are widespread in diagnostic laboratories [5], [6]. In addition, colorimetric dipstick tests like Siemens' Multistix 10SG are preferred for identifying abnormalities in urine, such as elevated concentration of protein, creatinine, or altered pH [7]. While these methods are effective, they require skilled personnel, controlled environments, and, most importantly, financial limits and are not ideal for widespread or remote screening applications.

The proposed method introduces a portable, non-invasive approach for early and precautionary detection of elevated toxin levels, specifically Albumin and Creatinine in urine samples. The proposed system consists of a vial that contains a reagent within turn, reacts with the sample of urine and produces diverse colours, i.e., different variations of colour gradients, which depend upon the concentration of Albumin and Creatinine. An Android application developed using Android Studio, Kotlin, and OpenCV, captures the image of the vial and records the aforementioned colour gradients or the RGB colour values. These values are then matched with pre-recorded data sets, patterns, and equations (all derived from the data sets) to estimate the concentration of Albumin and creatinine, and then it shows the metrics on the app screen, indicating whether medical attention is recommended. Firebase is used for secure user authentication and optional cloud-based storage of test data. This cost-effective and user-friendly approach bridges the gap in accessibility and provides rapid screening, especially beneficial for remote or resource-limited sections of the population [8], [9].

The overall flow of the work has been depicted in [Figure 1](#).

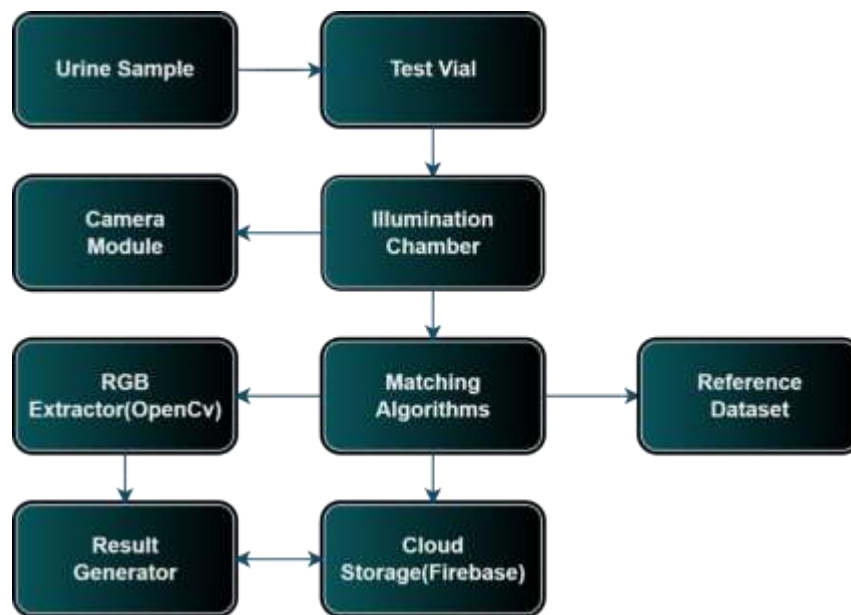


Fig. 1 Block diagram of the proposed urine-based blood toxin detection prototype

Every part of the diagram is described in the following subsections.



- **Urine Sample**

A urine sample is collected from the patient, and it is placed inside a designated vial; the vial contains a reagent that is sensitive to Albumin and Creatinine, which gives a color proportional to their concentration.

- **Test Vial**

The test vial contains the above-mentioned reagent and gives color gradient patterns based on the concentration of the Albumin and Creatinine, the intensity, color gradients, and shade. This color is the basis of detection in the next steps [8][5].

- **Camera Module (Android Phone)**

An Android device equipped with a macro lens is placed above the vial and uses our Android app using OpenCV to extract and analyze the color gradients produced by the vial. The camera is essential for real-time detection and provides a non-invasive application.

- **Illumination Chamber**

Inside the hardware prototype, this particular area is surrounded by multiple LED light strips and is covered by a polycarbonate plastic layer to diffuse the lighting and create a consistent lighting environment for the vial to ensure no sharp reflection/shadows occur throughout the process.

- **RGB Extractor (OpenCV)**

The Android application uses OpenCV and uses the live feed to extract the color gradients by properly separating the vial from the background. Then the values are averaged to be further used by the algorithm in the next step [10].

- **Matching Algorithm**

The extracted RGB values are matched within a pre-recorded referenced dataset, where the instantaneous values are checked for the range of results; if they fall within it also uses a few equations that were derived using the referenced dataset patterns. Hence, the equivalent concentration was found using the gradient values.

- **Reference Dataset**

This dataset is prepared by combining RGB and HSL/V values by doing a series of tests upon samples of various concentrations, and then they are placed to

provide equivalent RGB values for each concentration metric for Albumin and Creatinine ([Table 4](#), [Table 8](#)).

- **Result Generator**

The app analyses the extracted RGB values from the test samples and uses the matching algorithm to extract the approximate concentration of it using the reference dataset, then it concludes whether the concentration of Albumin and Creatinine is normal, elevated, or critical. The immediate result can be used by the user to determine whether physician consultation is required or not.

- **Cloud Storage**

The application stores the results and patients' details in online cloud storage (Firebase), which helps with the long-term tracking and remote access [\[11\]](#).

2 Literature Review

One study investigated the pre-symptomatic detection of the disease, comparing certain apps. The app is designed to recognize early symptoms of heightened blood toxicity and, therefore, potential kidney disease. It provides low-cost, user-friendly prototypes for rapid poster inspections and assists in addressing the challenges which are related to accessibility of diagnostics and provokes urgent medical attention in risky behavior groups. The instrument was also considered appropriate for use as a mass-screening tool because of the low level of training required and its compatibility with smartphone technology [\[12\]](#). A second study looks at how mobile and digital health technology can be employed for non-invasive kidney disease detection. It shows the advantages of a biosensor and a mobile app to find constituents such as albumin and creatinine. The authors focused on the fact that mixing both existing techniques and newer diagnostic tools can close the gap in terms of accessibility, specifically in rural and underserved areas. The work will also help with digital health, especially in the youth; it can also be used as an early detector of risk in the future, and will play a fundamental role in kidney care diagnostics, being less centralized [\[13\]](#).

In this proposal, a multifunctional and small enough urine analysis system is given for finding kidney disease markers like microalbumin and creatinine. The method uses Europium as a fluorescent indicator and colorimetric test strips for consistent measurement of the biomarker's concentration. It works well for repeated field application and shows significant opportunities for diagnosis with low resource requirements. The prototype was tested against a number of small test cases and gave good results [\[14\]](#). In this work, a portable device for detecting early kidney injury through Albumin-to-Creatinine Ratio (ACR) and NGAL markers is proposed. It is decided by colorimetric-based test methods,



image-based test methods, and their results. The device is cheap, rapid, and non-invasive, therefore suitable for remote or home-based requirements. It can greatly reduce dependency on lab infrastructure for early detection procedures and practices [15]. In machine learning research, clinical parameters were used to generate prediction models of CKD. Decision trees and random forest techniques are also used and feature an accuracy of over 99%. The work shows how AI can improve early diagnosis and allow for calculating CKD problems in large populations. Feature selection was also done to make sure that the model is more efficient by maintaining high performance while simultaneously reducing its input [16]. This review is about novel biomarkers (like KIM - 1, NGAL, and cystatin C) in the context of nephrotoxicity. These markers can detect kidney injuries before creatinine-based tests do. The work suggests that these markers should be added to current diagnostic systems to get better results. It adds that these markers need to be adopted pending broader clinical validation and the development of standard assays [17].

The present study details a sensor platform based on MIPs and multianalyte monitoring for CKD. There is simultaneous, selective, and multiplexed detection of the biomarkers by this point-of-care system. It is small enough to be portable and can be applied at home and on an individual basis, offering the potential for personal health monitoring. This technique is inspired by the mechanism of antibody-antigen interaction and is stable and cost-effective over biological receptors [18]. In this work, we report SERS detection of uremic toxins at trace levels via surface-enhanced Raman scattering (SERS) with plasmonic nanoparticles. It employs a commodity microfluidic chamber operated with microliters of urine, which yields high sensitivity and low resources. If it did, that could pave the way for an advanced kidney diagnostic in the pocket-sized spectrum. And it is extendable for use with smartphones or wearables to monitor in real time [19].

A summary of predictive artificial intelligence models found that AI can have a marked impact on early detection of CKD. The work states that machine learning and deep learning algorithms can greatly improve on a normal diagnostic test. It also shows how such models can be employed for monitoring and planning of treatment for a particular individual. In addition to big data processing and pattern recognition without human intervention, AI could help to verify records that may be overlooked during human observations [20]. TRACE is an AI model built with transformers that can predict early CKD from the medical history of patients. It uses deep learning and autoencoders to use visits and symptoms as input. The resultant model got a 2% increase in performance compared to existing models, which indicates that it can be applied to general diseases also. It also gives a case-by-case visual representation that talks about each prediction, improving the trustworthiness of the model [21]. STRAFE is their second-best model, but instead of directly focusing on CKD 3 to 5 progression, it uses lifetime modeling. It was tested with data from over 130,000 patients and bested some of the survival models like Cox regression. This structure helps physicians to predict disease progression and possibly schedule early intervention to improve it. This works by taking

into account past visits and progression patterns or tendencies of the particular patient. Hence, predicting the timeline and associated risk [22]. An understandable CKD prediction model was developed by Pedro Moreno-Sanchez, using XGBoost. It gives importance to interpretability and clinical significance by identifying & highlighting important features such as packed cell volume and hypertension. The model is clinically viable and can be applied in real-world medical healthcare, with a 99% accuracy. The SHAP values also help clinicians derive the impact of each attribute on the prediction [23].

This work presents a deep learning group or cluster combining EfficientNetV2, MobileNetV2, InceptionNetV2, and models for efficient visual processing for inputs. It achieved 96% accuracy for CKD detection and used detailed and structured AI algorithms to be understandable to clinicians. The model is particularly useful in classifying early vs. later CKD stages using imaging and clinical datasets [24]. Researchers used machine learning algorithms on data from over 11,000 patients in clinical testing to derive likely to happen kidney related disease in diabetic patients. Neural networks and regression models achieved AUC values between 0.81 and 0.84. The results can help target measures early and reduce the chances of ESRD progression. Albumin, Uria, Uric acid, and serum Creatinine were identified as the most critical biomarker predictors [25]. A deep learning approach was proposed to predict CKD 6 to 12 months before clinical detection of the beginning. Combining CNN, LSTM, and BLSTM models with multiple helper functions, the method achieved 98% accuracy. It shows the possibility for use in early detection in hospital EHR systems and mobile health tools. The system was tested on proper CKD datasets and supports real-time or near-instant predictions [26].

3 Methodology

The methodology used in this study was focused on the detection of toxin levels present in the human body by measuring the concentration of biomarkers by taking a urine sample and using a point-of-care (POC) device based on image processing. The process involved the creation of reference datasets for both Albumin and Creatinine using standard biochemical reactions that produce colour changes depending on the concentration of Albumin and Creatinine. These colour gradients were then analysed by an image scanning device and converted into distinct RGB and other parameter values, which are then used in the mobile application to estimate real-time concentrations from test samples by comparing the extracted colour gradient value to those present in the reference dataset. The complete methodology is shown in [Figure 2](#).

3.1 The proposed urine-based blood toxin detection prototype

The block diagram shows the workflow of the urine-based toxin detection prototype, designed to estimate concentration levels of Albumin and creatinine, which are important indicators of potential kidney issues. The process starts



with collecting a urine sample into a vial containing chemical reagents that react with biomarkers, causing a colour change that depends on toxin concentration. This reaction results in a colour that forms the basis of detection. After the reaction, the vial is placed in a custom illumination chamber with diffused LED lighting for uniform brightness and minimal shiny edges on the vial. An Android phone camera is placed above the vial. It then captures a focused image. Using OpenCV, the system extracts RGB values from a specific region of the image(vial), showing the reaction's colour intensity.

These RGB values are compared to a pre-recorded & standard reference dataset containing colour gradient parameters (R, G, B, H, S, L/V) to known Albumin and Creatinine concentrations. A matching algorithm estimates toxin levels and categorizes results (e.g., normal or elevated), which are displayed in the app. Data can also be securely uploaded to Firebase Cloud Storage. This system is fast, affordable, and accessible, ideal for early detection & safeguard, especially in rural or under-resourced areas without easy access to labs.

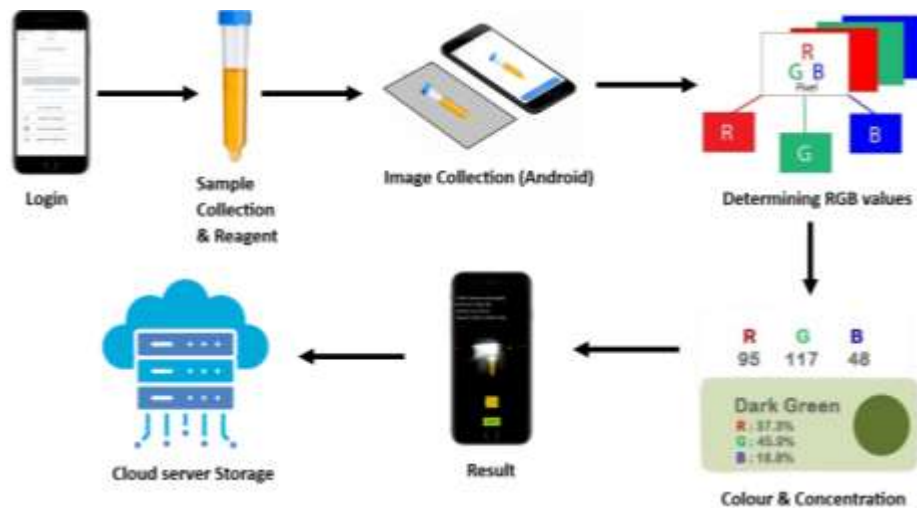


Fig.2 Process flow of the prototype hardware and application software

- **Login:** The user begins by opening the mobile application and securely logging in using their registered credentials. Firebase Authentication is used to verify the user and ensure that all subsequent data is associated with their profile for privacy and tracking purposes.
- **Sample Collection & Reagent:** The user collects a fresh urine sample and adds it to a specially designed test vial pre-filled with chemical reagents. These reagents are sensitive to biomarkers like Albumin and Creatinine and will trigger a visible color change based on the concentration present in the urine.

- **Image Collection:** The user places the vial just below the phone's camera inside the illumination chamber, where controlled and diffused lighting is incorporated. The camera acquires a high-resolution image for later steps.
- **Determining RGB values:** After acquiring the image, the application uses OpenCV to separate the region of interest (ROI) from the background, i.e., the vial; then the application extracts the RGB color gradient values from that region.
- **Color & Concentration:** Thus, the resultant acquired RGB values are compared with the prerecorded datasets of known concentrations of Albumin and Creatinine and their respective RGB values, so the application maps the result to the approximate concentration values.
- **Result:** Based on the concentration, the application categorizes it into "Normal", "Elevated" and "Critical" offering the user to know their real-time approximate health status (Albumin and Creatinine Concentration).
- **Cloud server storage:** Furthermore, the results, user details, and other related metadata (time, size, captured images) are stored optionally. These can be accessed after authentication and help with remote access to the data.

3.2 The Application Model

This application is designed for precautionary testing for blood toxins (Albumin & Creatinine) in the human body, which in turn helps to identify multiple kidney disorders in advance. This provides a user-friendly, cost-effective, non-invasive solution so that it can be used by a wider population, and one can know whether he/she needs further medical attention in this regard. In most of the cases, the higher/ critical concentration is caught after the patient has already suffered an ailment or is going through renal disorders. However, if such complications can be detected in advance, this application serves precisely that purpose. The following section outlines the existing problems and how this solution addresses them. Previously, blood toxin testing involved the use of blood samples, which posed several challenges. Drawing blood could sometimes lead to changes in toxin concentration, and many patients, especially the elderly, were reluctant to undergo blood collection. To address these issues, this prototype is designed to use urine as a non-invasive alternative to blood for toxin detection.

3.3 Reagent creation, testing, and deriving the reference dataset

The process of preparing reagents is discussed in three parts, described below. A prepared albumin standard of different concentrations is shown in [Figure 3](#). Dilution of Albumin Standard (100mg/dl) is also shown in [Table 1](#). The reagent and sample volume for OD-based albumin detection in 96-well Plates is reported in [Table 2](#). The final Albumin concentrations used in the OD-based



method (A620 Readings) are shown in [Table 3](#). Diluted creatinine of different standard concentrations is shown in [Figure 4](#).

3.3.1 Albumin Detection - Reagent Preparation & Dataset Generation:

Here is a list of some materials that have been used in the experiment. Bromocresol Green (BCG) – 0.35 mM, Succinic Acid – 100 mM, Albumin Primary Standard – 20 mg/ml, MilliQ Water, 96-well Microplate, Microplate Reader (OD-based detection), and Image Scanning Device (for RGB extraction) were used during the experimental study. To prepare a 0.35 mM BCG solution, 100 mM Succinic Acid is added to adjust the pH to 4.0. Then, the BCG solution is diluted to a final concentration of 0.15 mM (pH 4.0), which is used as the reagent. A 20 mg/ml Albumin standard is diluted with MilliQ water to make a 100 mg/dl stock. Next, further dilution is done to prepare the following concentrations ([Table 1](#)).

Table 1 Dilution of Albumin Standard (100mg/dl)

SL no.	Albumin Standard (ml)	Water (ml)	Final vol. (ml)	Diluted Std Conc.(mg/dl)
1	10	0	10	100
2	5	5	10	50
3	3	7	10	30
4	1.6	8.4	10	16
5	1	9	10	10
6	0.6	9.4	10	6
7	0.5	9.5	10	5
8	0.1	9.9	10	1

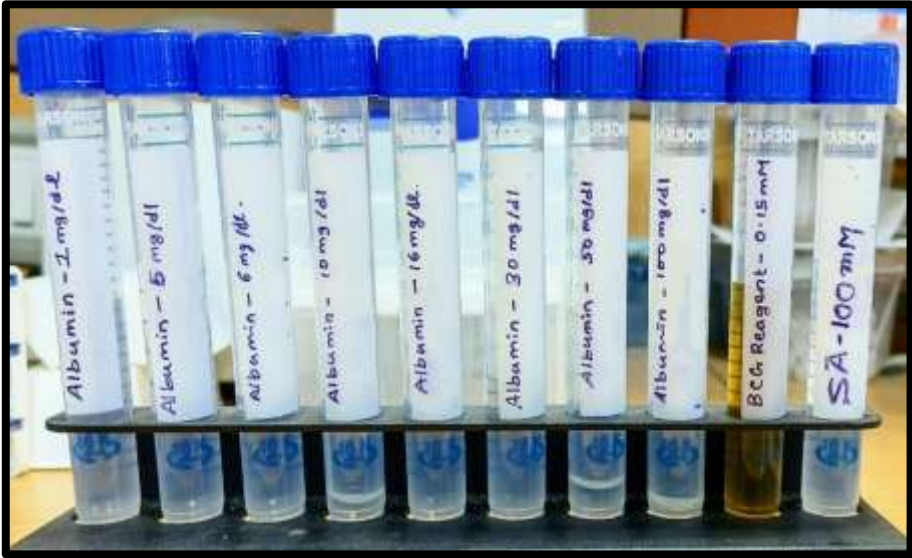


Fig.3 Prepared albumin standard of different concentrations

OD-Based Method Steps: Here is a description of the concentration preparation with the OD-based method. A 100 μ L of BCG reagent is transferred into each well of a 96-well plate. A 100 μ L of each Albumin standard solution is added to the respective wells. The solution is mixed gently. The final volume per well is 200 μ L. The final concentrations in wells are reported as 0.5, 2.5, 5, 15, 25, 50 mg/dl. Then the solution needs to be kept in incubation for 1 minute at room temperature. Then the final measurement is done on absorbance at 620 nm (A620) using endpoint mode (as shown in [Table 3](#)).

Table 2 Reagent and Sample Volume for OD-Based Albumin Detection in 96-Well Plate

Initial Diluted Alb Std Conc. (mg/dl)	Diluted Alb Std (μ l)	Water (μ l)	BCG Reagent (μ l)	Final Volume (μ l)	Final Diluted Alb Std Conc. (mg/dl)
0 (Blank)	0	100	100	200	0
1	100	0	100	200	0.5
5	100	0	100	200	2.5
6	100	0	100	200	3



10	100	0	100	200	5
16	100	0	100	200	8
30	100	0	100	200	15
50	100	0	100	200	25
100	100	0	100	200	50

Table 3 Final Albumin Concentrations Used in OD-Based Method (A620 Readings)

Alb Conc. (mg/dl)	OD ₁	OD ₂	OD ₃	OD ₄	OD ₅	OD _{AVG}	Std Dev.
0.5	0	-0.007	0	0.001	0.003	-0.001	0.003
2.5	0.004	0.003	0	0.004	0.003	0.003	0.001
5	0.017	0.019	0.016	0.016	0.019	0.017	0.001
15	0.122	0.129	0.121	0.118	0.126	0.123	0.004
25	0.193	0.214	0.198	0.216	0.208	0.206	0.009
50	0.491	0.491	0.496	0.526	0.514	0.504	0.014

Image Scanning Method Steps: For this step, 100 μ L of BCG reagent was first transferred into the designated wells of the microplate. Following this, 100 μ L of Albumin standards, with concentrations of 6, 10, 16, 30, 50, and 100 mg/dl, were added to the respective BCG reagent wells. The samples were gently mixed for uniform composition throughout, and finally, samples of volume 200 μ L were prepared. After mixing them, the final concentrations are 3, 5, 8, 15, 25, and 50 mg/dl. Each individual test was carefully performed in duplicate and repeated three times to maintain its consistency. Then the prepared solutions were left to rest at room temperature so that the reaction could complete and the reaction stabilizes. Once the reactions were complete, the wells were loaded into the image scanner. Thus, the acquired results, specifically the RGB values for each concentration, were recorded as provided in [Table 4](#).

The dataset includes RGB values and combined RGB (R+G+B) intensity values for the samples ranging from 0 mg/dL to 50 mg/dL. For each concentration, the sample was scanned 6 times to extract the average (Avg) metric and the standard deviation (STDV). The prototype used to derive the dataset is the earlier iteration of this prototype, developed by CDAC.

Table 4 R, G, B values (Albumin) extracted by the standard device

Alb Conc.	R	G	B	R+G+B
50mg/dl	49.507	72.913	68.609	191.029
	30.290	59.710	49.725	139.725
	30.928	60.174	50.261	141.362
	30.333	59.942	49.420	139.696
	25.855	57.449	46.522	129.826
	31.507	60.783	50.232	142.522
	AVG	33.070	61.829	52.461
STDV	8.300	5.548	8.032	21.867
15mg/dl	66.188	72.275	38.551	177.014
	64.942	70.855	33.217	169.014
	64.609	70.072	34.058	168.739
	64.348	72.014	35.478	171.841
	66.058	72.536	35.536	174.130
	65.391	70.783	34.594	170.768
	AVG	65.256	71.423	35.239
STDV	0.758	0.987	1.844	3.185
8mg/dl	73.797	71.072	29.725	174.594
	75.203	74.145	30.029	179.377
	75.275	73.203	29.971	178.449
	74.014	70.971	27.725	172.710
	77.348	76.145	31.565	185.058
	74.681	72.145	26.913	173.739
	AVG	75.053	72.947	29.321
STDV	1.275	1.989	1.701	4.624
5mg/dl	81.232	74.290	29.493	185.014
	81.029	74.333	27.507	182.870
	78.435	71.391	26.029	175.855
	81.913	74.493	29.217	185.623
	82.594	75.826	27.942	186.362
	80.406	73.855	28.855	183.116
	AVG	80.935	74.031	28.174



	STDV	1.438	1.456	1.295	3.826
3mg/dl		82.957	74.275	28.754	185.986
		81.043	71.870	25.377	178.290
		78.768	71.435	24.551	174.754
		83.449	75.261	29.594	188.304
		81.928	72.478	26.174	180.580
		82.464	73.261	25.536	181.261
	AVG	81.768	73.097	26.664	181.529
	STDV	1.690	1.466	2.029	4.963
omg/dl		80.188	69.913	20.623	170.725
		81.536	71.638	22.377	175.551
		85.580	74.188	26.319	186.087
		82.507	72.029	23.812	178.348
		82.174	71.536	20.551	174.261
		86.246	74.217	24.000	184.464
	AVG	83.039	72.254	22.947	178.239
	STDV	2.373	1.675	2.222	5.998

These findings suggest that acquired RGB values are quite unique with respect to individual different concentration present in the samples the unique combination of RGB values can serve as a consistent indicator for toxin levels. By Observing the data in some places like low to mid concentration range the difference between the readings of different concentration are quite distinguishable across repeated tests. This consistency and accuracy of the RGB values and toxin concentration further supports the effectiveness of image-based analysis for early screening of kidney related toxins and also shows system's potential to become a practical approach.

3.3.2 Creatinine Detection:

Reagent Preparation & Dataset Generation: Picric Acid at a concentration of 1 mmol, Sodium Hydroxide at 0.29 mol/L, Creatinine Primary Standard at 36 mg/ml, MilliQ Water, a 96-well Microplate, and a Microplate Reader were used for this procedure. For the preparation of the standard creatinine solution, 36 mg/ml creatinine was first diluted to make an 1800 mg/dl stock solution. Then, a 1350 mg/dl working solution was prepared (as per [Table 5](#)). This working solution was further diluted to generate the following standard concentrations (as per [Table 6](#)) ([Figure 4](#)).



Fig. 4 Diluted creatinine of different standard concentrations

Table 5 Dilution of Creatinine Standard (1350 mg/dl) to Prepare Working Solutions

Sl no.	Creatinine standard (ml)	Water (ml)	Final vol. (ml)	Diluted creatinine std conc. (mg/dl)
1	2	0	2	1350
2	1.33	0.67	2	900
3	1.067	0.933	2	720
4	0.533	1.467	2	360
5	0.267	1.733	2	180
6	0.2	1.8	2	135
7	0.133	1.867	2	90
8	0.067	1.933	2	45



Table 6 Reagent and Sample Volume Used for OD-Based Creatinine Detection

Initial Diluted Crn Std Conc. (mg/dl)	Diluted Crn Std(μl)	Water (μl)	NaOH (μl)	Picric Acid (μl)	Final Volume (μl)	Final Diluted Crn Std Conc.(mg/dl)
0 (Blank)	0	140	10	30	180	0
18	20	120	10	30	180	2
45	20	120	10	30	180	5
90	20	120	10	30	180	10
135	20	120	10	30	180	15
180	20	120	10	30	180	20
360	20	120	10	30	180	40
720	20	120	10	30	180	80
900	20	120	10	30	180	100
1350	20	120	10	30	180	150
1800	20	120	10	30	180	200

OD-Based Method Steps: Thirty microliters of picric acid were added to nine consecutive wells. Then, ten microliters of NaOH were added to each well. After that, 120 microliters of MilliQ water were added to each well. Next, 20 microliters of diluted creatinine standards were added to the respective wells. The solutions were gently mixed to prepare a final volume per well of 180 μL. The final concentrations were 2, 5, 10, 15, 20, 40, 80, 100, and 150 mg/dl. The plate was incubated for 15 minutes at room temperature. Finally, the absorbance at 400 nm (A400) was measured, with the values given in [Table 7](#).

Table 7 Final Concentrations of Creatinine Used in the Image Scanning Method

OD-BLANK							
Crn Conc. (mg/dl)	OD ₁	OD ₂	OD ₃	OD ₄	OD ₅	OD _{AVG}	Std Dev.
2	-0.004	0.001	-0.003	0.003	-0.025	-0.006	0.011

5	0.013	0.006	0.004	0.004	0.013	0.008	0.005
10	0.027	0.017	-0.009	0.04	0.018	0.019	0.018
15	0.02	0.022	0.034	0.039	0.032	0.029	0.008
20	0.029	0.035	0.041	0.042	0.023	0.034	0.008
40	0.085	0.078	0.093	0.095	0.09	0.088	0.007
80	0.147	0.134	0.158	0.138	0.15	0.145	0.010
100	0.151	0.168	0.198	0.193	0.158	0.174	0.021
150	0.241	0.246	0.245	0.254	0.233	0.244	0.008

Image Scanning Method Steps: Thirty microliters of picric acid were added to each well. Then, ten microliters of NaOH were added to each well. After that, 120 microliters of water were added and mixed gently. Next, 20 microliters of creatinine standards ranging from 45 to 3600 mg/dl were added. The final concentrations in the wells were 5, 15, 20, 40, 100, 200, and 400 mg/dl. The plate was incubated for 15 minutes at room temperature. The plate was then loaded into the image scanner. Each well was selected and scanned. Finally, values for R, G, and B were extracted and reported in [Table 8](#).

Table 8 R, G, B values (Creatinine) extracted by a standard device

Crn Conc.	R	G	B	R+G+B
40mg/ dl	101.710	85.377	61.000	248.087
	108.000	88.014	60.029	256.043
	108.333	90.014	62.406	260.754
	109.551	91.261	64.855	265.667
	108.391	89.246	63.130	260.768
	107.058	88.942	60.522	256.522
	93.609	76.884	52.609	223.101
AVG	105.236	87.106	60.650	252.992
STD DEV	5.726	4.864	3.918	14.263
20mg/ dl	104.232	92.797	74.899	271.928
	107.841	94.362	72.957	275.159
	107.681	93.986	71.725	273.391
	108.957	96.638	74.841	280.435
	107.696	95.681	75.986	279.362
	105.058	92.739	71.884	269.681



		96.116	84.290	66.014	246.420
	AVG	105.369	92.928	72.615	270.911
	STD DEV	4.411	4.066	3.330	11.465
15mg/dl		101.348	89.768	71.145	262.261
		105.623	94.725	73.986	274.333
		106.667	95.725	75.696	278.087
		104.725	92.667	72.797	270.188
		105.232	94.174	74.942	274.348
		103.942	93.609	75.101	272.652
		94.087	83.957	67.826	245.870
	AVG	103.089	92.089	73.070	268.248
	STD DEV	4.307	4.055	2.788	11.037
5mg/dl		107.014	107.014	89.246	297.304
		109.928	109.928	89.159	301.681
		110.478	110.478	91.522	305.986
		107.725	107.725	87.130	295.609
		107.681	107.681	88.130	297.000
		107.304	107.304	88.957	298.870
		97.652	97.652	82.681	273.304
	AVG	106.826	106.826	88.118	295.679
	STD DEV	4.261	4.261	2.744	10.468

3.4 Prototype Structure (Hardware Components)

As already discussed, the urine will be used to obtain the required results. The prototype uses a particular vial that will contain a reactant sensitive to Albumin & Creatinine. So, it will give different colors to different concentrations of albumin. This vial would be placed in a particular position. The portion of the prototype that will contain this vial will be perfectly illuminated with surrounding diffused light, which will help to identify the colors more accurately. Right above the vial, the image sensor will be placed (in this iteration of the prototype, an Android device is used due to its low focal length and good auto-focusing). For capturing and computing the color gradients of the provided vial and Android, this device will be used with Android version 8.2 or higher. This will also contain the developed software built within it. The other portion of the prototype will contain batteries and switches to deliver the required energy to illuminate the vial's environment. Please refer to [Figure 5](#) for the detailed prototype.

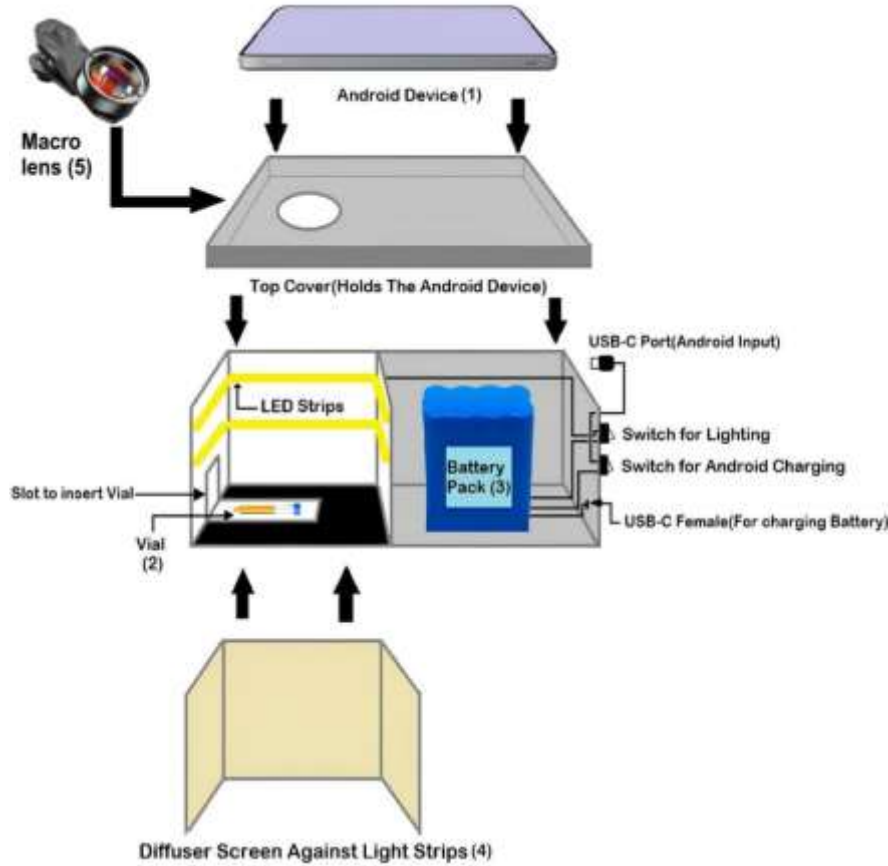


Fig. 5 Structural prototype of the hardware case

- **Android Device:** The primary image-capturing and processing device (Android version 8.0 or newer).
- **Vial:** The vial contains the reagent, as previously discussed, and is inserted into the hardware through a designated slot.
- **Battery Pack:** The battery pack powers both the Android device and the LED strips. Charging the battery pack alone is sufficient; there is no need to charge the Android device and the battery pack separately. (Specification: 12V, Rechargeable 5200mAh, Lithium-Ion Battery)
- **Diffuser Screen:** The diffuser screen layer diffuses the sharp light given off by the LED strips, preventing harsh or sharp shadows over the vial (which is also quite reflective) and maintaining uniform light intensity. (Specification: Polycarbonate LED Diffuser).
- **Macro Lens:** The macro lens decreases the focal length of the Android device's camera; typically, its conventional focal length is way over 25



Centimeters, therefore, this lens will allow it to capture images with improved focus and clarity. (Specification: Aperture: F1.8, Minimum Focal Length: 33 mm, Diameter: 1.57 inches).

3.5 Android Application

Below each component is explained in detail, its technical properties, and the stacks used. A few screenshots of the application are also given in [Figure 6](#).

3.5.1 Firebase-based Authentication System using Android Studio

The authentication part of the prototype was made using Android Studio by using Firebase Services [\[27\]](#), [\[28\]](#) to give secure and easy user login and registration processes. This makes sure that only the allowed user can access the toxin detection application and its data, also providing user-specific separation. The user login records or logs are stored in Firebase, as shown in [Figure 7](#).

Dependencies and Libraries Used:

- Firebase Authentication SDK: Open-source solution that provides backend services and functions for authenticating users using proper credentials.
- Google Services JSON: Connects the Android application to the Firebase project and validates that the application really has a valid instance of that particular project each time an auth request is made.
- AndroidX Libraries: Support modern Android development tools like UI management and thread handling.
- Material Design Components: Used for the UI components inside the screen, like background, buttons, text fields, and stuff.
- Firebase Core Libraries: Handle authentication and user session, i.e., if verifies whether a valid user session is present or not.

Application Flow and Functionality:

1. Authentication Initialization: A FirebaseAuth instance is created using FirebaseAuth.getInstance() to enable authentication operations throughout the application.
2. User Validation at Startup: On application start, the current user session is checked. If it finds a valid session after confirming, the user will be redirected directly to the MainActivity, or else the user will be moved to the Login screen.
3. User Login Screen: The user can log in by entering “Username” and their “Password”.

4. Login Authentication: When the user clicks the login button, the application authenticates the user if it returns “Success”; thus, the screen will switch to “MainActivity” and if it fails, an error toast pops up [27], [28], [29].
5. Registration Screen Navigation: Users without an account can tap "Register Now" to navigate to the Registration activity.
6. User Registration Process: On the registration screen, users enter their email and password, then validation takes place, and after successful validation, a new user account is created.
7. Registration Success and Redirect: After successful registration, go to STEP 3.
8. Progress Indication: A progress bar is shown during the process and registration processes to provide visual feedback during background operations.
9. Logout Functionality: When the “Log Out” button is pressed it will end the user session by executing FirebaseAuth.getInstance(). signOut () then STEP 3.
10. Error Handling and Feedback: Proper pop up/ toast messages are displayed in the user login page informing the user about the error e.g. incorrect credentials or network failures.
11. Back Stack Management (back navigation): After successful login or registration, finish() to prevent unintended or false back navigation.

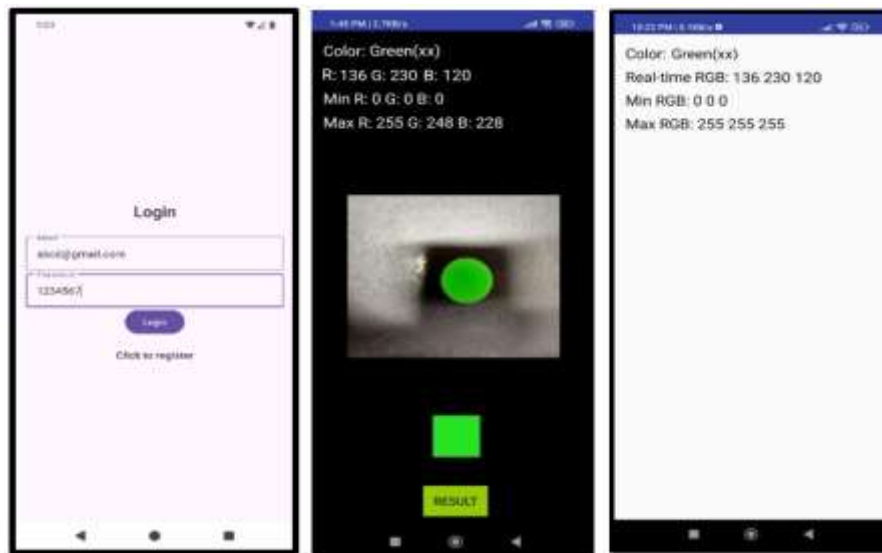


Fig.6 Some Screenshots from the Android App

(a). Page 1: login & registration (user authentication); (b): Page 2: R, G & B value detection (c): Page 3: Result generation



Fig. 7 User login record in Firebase (Firebase project console web)

3.5.2 OpenCV-based RGB Detection using Android Studio

The software component of the prototype was developed using Android Studio, with the integration of the OpenCV library to perform real-time color-based image processing [30]. The aim was to extract the RGB values of a urine reagent sample through a mobile camera and use those values for toxin estimation based on pre-calibrated reference datasets.

Dependencies and Libraries Used:

- OpenCV Android SDK: Enables real-time image processing with support for native C++ and Java bindings.
- CameraBridgeViewBase: OpenCV's high-level camera interface for Android that allows frame-by-frame processing via CvCameraViewListener2 [31].
- Imgproc (Image Processing module): Used for tasks such as color space conversion, affine transformations, and drawing contours or rectangles.
- Core (OpenCV core module): Provides functions for matrix manipulation and basic computer vision operations like summing pixels, scalar operations, and contour drawing.

Application Flow and Functionality:

1. Camera Initialization and Frame Capture: The camera feed gets displayed while using CameraBridgeViewBase, and every frame is transferred to the onCameraFrame () method. OpenCV processes the frame in RGBA format using Mat (Matrix) objects.

2. **Region of Interest (ROI) and Rotation:** For better visuals, the camera frame is rotated by 90 degrees counterclockwise using `getRotationMatrix2D()` and `warpAffine()` functions. A rectangular region (ROI) is defined inside the camera feed at its center ($x/2$, $y/2$) to separate the vial from the background and only process the color gradients present in the vial.
3. **Color Conversion and RGB Extraction:** The submatrix within the ROI is taken and converted from RGB to HSV color space using `Imgproc.cvtColor()`. The HSV values are combined and averaged over the entire region to create a single majority color, then converted back to RGBA using a helper function. These RGB values represent the detected color from the test sample [32].
4. **Tracking Min/Max RGB Values:** The application continuously extracts the RGB values from the ROI, i.e., the vial, and also records the minimum and maximum RGB values to create a range of metrics for future result mapping.
5. **Contour Detection and Display:** A custom color blob detector (HSV thresholding) is used to identify and draw contours within the region. `Imgproc.drawContours()` visually outlines detected blobs; these outlines ensure that the ROI being detected is actually correct, and the user can check it in real time if it goes wrong. Then the test can be reinitiated.
6. **Color Classification:** A predefined set of RGB value ranges is already defined by known color categories (e.g., red, green, unknown) for user reference. The detected RGB values are compared against these ranges to determine a probable concentration range (e.g., "Unknown (5–6 mg/dl)"). Thus, providing real-time instant result generation
7. **User Interface & Result Handling:** After getting the RGB values of the selected color, it also gets the highest and lowest RGB values to get the deviation range. After that, the result button gets active; when pressed, it will take the user to the result screen.
8. **Code Structure and Threading:** Android's Handler & Runnable interfaces are used to add a delay for the camera view. `runOnUiThread()` - this delay helps the UI to get updated (like RGB display and ImageView color rendering) in time because it's a heavy process in comparison to convention android processes; if not properly handled will cause issues. Also safely handles the camera feed by using separate threads; the delay ensures that the application doesn't crash under load.

This approach integrates real-time image processing with a user-friendly UI, enabling quicker and non-invasive estimation of toxin levels present in a urine sample. The use of OpenCV helps with handling frame operations with accurate



color analysis, while Android's architecture enables smooth user interaction and multi-threaded execution for the application.

4 Results & Discussion

The prototype consists of an illumination chamber where the user inserts a vial containing the urine sample & reagent. Next, the user logs in to the application inside the Android device using his/her registered user ID and password. Then, inside the application, the camera captures images of the vial placed inside, where it detects the region of interest by separating the vial from the black base. RGB values and minimum ~ maximum ranges are extracted from the ROI, and then the acquired results are mapped with the prerecorded data to find the approximate concentration of Albumin and Creatinine inside the vial (Urine Sample). After 15 seconds, the above-mentioned results are generated, and the application navigates to the result screen where all the above-mentioned resultant metrics are displayed.

This application aims to design an affordable, accessible, and user-friendly prototype for estimating blood toxin levels, specifically Albumin and Creatinine, through non-invasive urine sample analysis. Using mobile-based image processing and Firebase authentication, the system is intended for early detection of kidney-related health issues, particularly in underserved populations with limited access to diagnostic facilities. Preliminary testing showed an average accuracy of 97.51%, demonstrating strong potential for effective early screening. This section describes the testing results done on the samples provided by CDAC, Kolkata. [Figure 8](#) shows a tray with different Albumin concentrations of the samples provided for the testing.

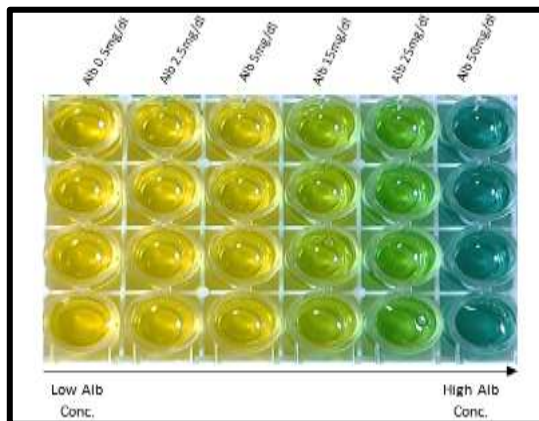


Fig. 8 Test samples (Albumin)

Evaluation Metrics

To measure the performance of the extracted RGB values against the reference dataset, we define Accuracy and Error as follows.

The formula for Accuracy is given by

$$\text{Error} = \left(\frac{|\Delta r|}{R} + \frac{|\Delta g|}{G} + \frac{|\Delta b|}{B} \right) \div 3 \times 100\% , \text{ where:}$$

- **$\Delta r = |R-r|$, where R represents the average red gradient value from the reference dataset (Table 4), and r denotes the red gradient value extracted by the machine.**
- **$\Delta g = |G-g|$, where G represents the average green gradient value from the reference dataset (Table 4), and g denotes the green gradient value extracted by the machine.**
- **$\Delta b = |B-b|$, where B represents the average blue gradient value from the reference dataset (Table 4), and b denotes the blue gradient value extracted by the machine.**

Thus, the accuracy percentage is calculated as:

$$\text{Accuracy} = (100 - \text{Error}) \%$$

This formula provides a direct comparison between the mobile-generated results and the respective reference dataset values, and combines their effective deltas across all three RGB channels. By calculating how closely the measured color gradient values match the reference dataset ones. A higher accuracy value indicates a closer similarity with the reference dataset, signifying more reliable and consistent performance of the system. Conversely, a higher error value suggests a greater difference from the expected results, indicating potential inconsistencies or inaccuracies in the measurement process.

Two test results of different albumin concentrations, obtained with the developed software, are shown below [Figure 9(a) and Figure 9(b)]. The concentration of Albumin is defined as 5mg/dl and 15mg/dl, respectively, in both cases. Calculating the results on the full dataset, the average test result has been reported as 97.51%.

In Figure 9(a), the test results obtained by the application are r:81 g:73 b:30. Originally, from the test results of records (Table 4), it was R:80 G:74 B:28. So, the calculated error is:

$$(((1/80) + (1/74) + (2/28))/3) * 100\% = 3.24\%. \text{ Therefore, the Accuracy is } 96.76\%$$



In Figure 9(b), the test results obtained by the application are r:66 g:70 b:35. Originally, from the test results of records (Table 4), it was R:65 G:71 B:35. So, the calculated error is:

$((1/65) + (1/71) + (0/35))/3 * 100\% = 0.97\%$. Therefore, the accuracy is 99.03%.

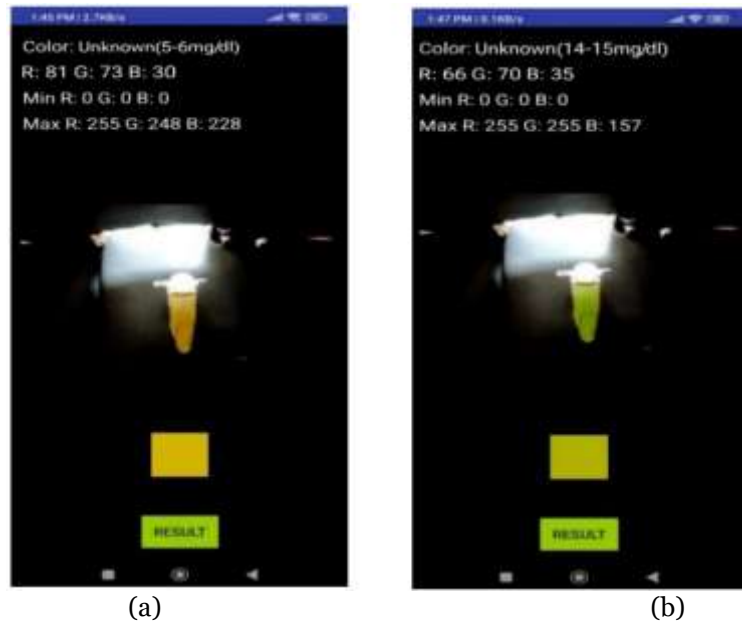


Fig. 9 RGB values & Albumin concentration obtained from application (a). 5mg/dl; (b). 15mg/dl

Table 9 Performance Evaluation of RGB-Based Albumin Concentration Estimation Using the Reference Dataset

Sl. No.	Expected Concentration (mg/dL)	R	G	B	r	g	b	Accuracy (%)
1	50	33	62	52	34	61	51	97.82
2	15	65	71	35	66	70	34	97.08
3	8	75	73	29	74	72	30	97.69
4	5	81	74	28	80	73	28	97.91
5	3	82	73	27	81	73	26	98.37
6	0	83	72	23	82	71	22	97.52
7	15	65	71	35	64	70	35	97.49
8	8	75	73	29	76	72	28	96.86
9	5	81	74	28	82	73	27	96.89
10	3	82	73	27	83	72	26	96.67

* R, G, and B represent the average reference RGB values (rounded to the nearest integer) obtained from the reference dataset, whereas r, g, and b represent the corresponding RGB values extracted by the proposed image-processing system.

Therefore, considering all twelve test cases (including the two representative test cases shown in [Figure 9](#) and the ten additional validation samples presented in [Table 9](#)), the average accuracy is calculated as:

$$\frac{(96.67)}{12} = 97.51\%$$

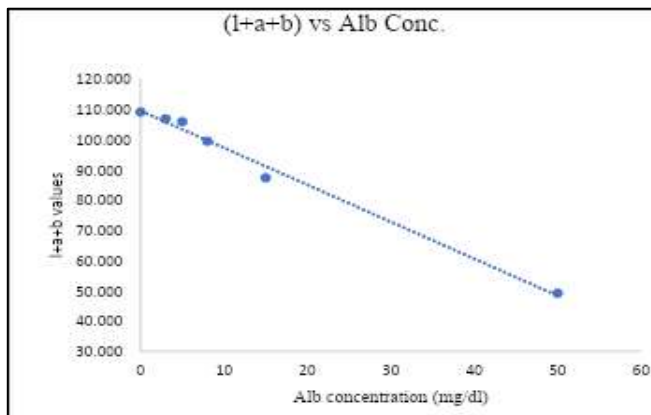
Hence, the average error is:

$$100 - 97.51 = 2.49\%$$

Therefore, Total approximate error is 2.1% $((3.24 + 0.97)/2)$ and results accuracy is 97.9% $((96.76 + 99.03)/2)$.

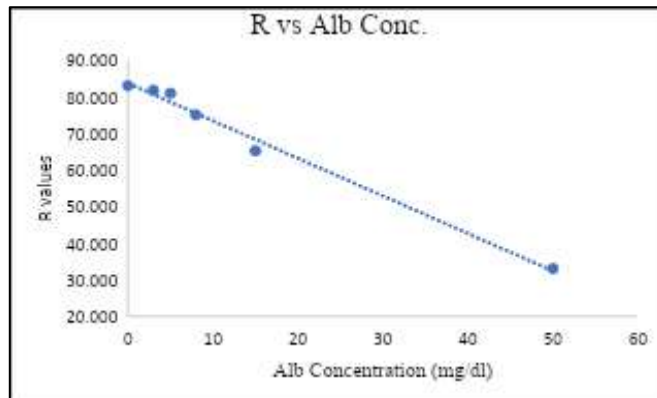
After plotting the value of R & B against albumin concentration, we get the following graphs; from here, the equations can be concluded.

The nature of Red & Blue color components with increasing concentration of Albumin is shown in [Figure 10](#) and [Figure 11](#).



Alb Conc.	0	3	5	8	15	50
R values	22.947	26.664	28.174	29.321	35.239	52.461

Fig. 10 Graph showing relation between R values and Albumin concentration



Alb Conc.	0	3	5	8	15	50
R values	22.947	26.664	28.174	29.321	35.239	52.461

Fig. 11 Graph showing relation between B values and Albumin concentration

Limitations & Future Development

- **Limited Test Cases:** Currently, only a small number of test cases have been used to train the algorithm for image processing. In the future, a larger dataset will be developed through extensive testing to improve the model's accuracy and to identify additional patterns that could enhance diagnostic insights.
- **Cloud Storage:** Currently, only user authentication credentials and data are stored in the cloud (Firebase). In the future, focusing on storing all generated results/ analysis along with patient records and medical history will be prioritized for better tracking.
- **User Interface:** Currently, the user interface is very basic and minimalistic, satisfying the functional requirements only; future updates will aim to make the interface better-looking and user-friendly.
- **Integration:** For now, the application is not a single whole system; the image processing, color gradient extractor, and result generation are placed in one part, and the other part includes user authentication and login. Future improvements will include a full-stack solution for every step in one application only.
- **Equation Implementation:** The formulas derived (refer to [Figure 10](#) and [Figure 11](#)), in the future, more such equations/ relations can be found with the expanding datasets and test cases

5 Conclusion

The intended and primary goal of this work was to make a low-cost, accessible and user-friendly prototype that can estimate blood toxin concentrations (more specifically Albumin, Creatinine & ACR), through non-invasive method by analyzing urine sample color gradient through adding chemical reagents to it, further utilizing mobile-based image processing application (OpenCV) with Firebase authentication, the system intends to support early detection of kidney-related health issue before-handed, especially people with limited access to diagnostic services. The preliminary results showed an average accuracy of 97.51%, demonstrating that the prototype is very promising for early screening and detection. Nevertheless, it is not yet perfect as there remains a prominent scope for further enhancement, especially through expanding the reference dataset by making it vaster and more varied, and identification of deeper patterns and relations to further increase the precision and accuracy in complex test cases of the OpenCV-based color detection system. With consistent enhancement in data handling, color analysis, and machine learning model implementation and optimization, the prototype can grow into a highly useful tool for real-world deployment and use cases. This prototype can serve as a reliable indicator for determining whether toxin concentrations are at alarming levels (not precise concentration when compared to conventional lab procedures). It is inexpensive, highly portable, and well-suited for use in rural areas where access to medical facilities is limited, making it ideal for large-scale deployment. Therefore, while the current version serves as a valid and functional proof-of-concept, future improvements & fixes are necessary to fully realize its clinical and practical potential.

Repositories

The GitHub link (OpenCV colour recognition):

<https://github.com/Alpharangerr/USHA-image-Detection-OpenCV>

and GitHub link (Firebase user authentication):

<https://github.com/Pritam1408/USHA75>

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Contribution of each Author

Mr. Adarsh Nandi and Mr. Pritam Sadhu carried out the implementation of the proposed methodology, conducted the experiments, and prepared the initial documentation. Dr. Nilanjana Dutta Roy prepared and revised the manuscript, coordinated the overall study, and served as the corresponding author. The research concept and methodology were conceived by Dr. Subhankar Mukherjee, Dr. Souvik Pal, and Mr. Shubhajit Bhar, who also provided technical guidance throughout the study. The experimental work was carried out at C-DAC Kolkata with the support of the research team. All authors reviewed, revised, and approved the final version of the manuscript.

Conflict of Interest Statement

The authors declare no conflict of interest.

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Mr. Shubhajit Bhar is a biotechnology researcher with over nine years of experience in biosensors, environmental biotechnology, water quality assessment, and public health research. He has worked with leading research organizations, including C-DAC Kolkata, ICMR-NICED, CSIR-CGCRI, and IIST Shibpur. His expertise spans biosensor development, environmental diagnostics, laboratory research, and analytical techniques. He is proficient in advanced laboratory instrumentation and experimental data analysis. His research interests focus on developing innovative diagnostic technologies for healthcare and environmental sustainability. He is dedicated to translating scientific research into practical, technology-driven solutions for real-world challenges.



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Dr. Souvik Pal is a researcher with extensive expertise in nanotechnology, biosensors, and point-of-care diagnostic systems. His research focuses on the development of nano-surface-based biosensors for clinical and environmental monitoring. He has significant experience in biofunctionalization, bioassay development, and electrochemical and optical sensing technologies. He has successfully



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translated laboratory research into portable diagnostic devices, including AquaSuruksa, e-GUNA, and ColoSENS. Dr. Pal has published over 35 peer-reviewed international research papers and holds three Indian patent applications. His interdisciplinary research and innovation continue to contribute to the advancement of healthcare and environmental diagnostic technologies.